



Effects of ayahuasca and its alkaloids on substance use disorders: an updated (2016–2020) systematic review of preclinical and human studies

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Abstract

Ayahuasca is a hallucinogenic/psychedelic traditionally used for ritual and therapeutic purposes. One such therapeutic use is related to Substance Use Disorders (SUDs). A previous systematic review of preclinical and human studies published until 2016 suggested that ayahuasca and its alkaloids have therapeutic effects in the treatment of SUDs. To conduct an update of this previous review. A systematic review of quantitative studies which analyzed the effects of ayahuasca and its alkaloids on drug use (primary outcome) and other measures (secondary outcomes) related to SUDs was conducted, including articles from 2016 to 2020. Nine studies (four preclinical, five observational) were included in the review. Preclinical studies in rodents reported reductions in amphetamine self-administration and anxiety, and in alcohol- and methylphenidate-induced conditioned place preference. Observational studies among healthy ritual ayahuasca users and patients with SUDs reported reductions in drug use, anxiety, and depression, and increases in quality of life and well-being. We replicated the findings of the previous review suggesting that ayahuasca and its alkaloids have therapeutic effects in the treatment of SUDs. However, translation of preclinical data to humans is limited, observational studies do not allow us to infer causality, and there is a lack of standardization on ayahuasca doses. Although promising, randomized, controlled trials are needed to better elucidate these results. The PROSPERO ID for this study is CRD42020192046.

Keywords Hallucinogens · Psychedelics · Ayahuasca · Dimethyltryptamine · Harmine · Harmaline · Substance use disorders

Introduction

Substance use disorders (SUDs) are among the most prevalent psychiatric disorders, being associated with individual (daily malfunctioning, less productive life) and public

(criminality, socioeconomic costs) health-related problems. SUDs are chronic disorders characterized by relapses, compulsion to seek and consume substances, loss of control of consumption, and the emergence of negative emotional states such as irritability and dysphoria. They are often associated with physical and/or other mental health disorders, such as anxiety and depression [1–5].

SUDs can be caused by the use of commonly licit (such as alcohol, tobacco, and prescription drugs) or illicit substances (such as stimulants, opioids, and cannabinoids), with varying degrees of negative consequences depending on the drug, social and demographic characteristics of the user, and/or availability [1–5]. Indeed, some researchers classify alcohol and tobacco together with heroin and crack-cocaine as the most harmful drugs for both the individual and society, whereas drugs such as cannabis and ecstasy show less harm [5]. Alcohol and tobacco are more readily available and are culturally accepted in most countries, which is

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reflected in higher harm and incidence of problematic use. Heroin and crack-cocaine, although less available and culturally accepted, are often associated with compulsive use and law and criminal problems [1, 4, 5].

Current treatments for SUDs include individual or group psychotherapy and/or psychiatric medications. Regarding pharmacological treatments, nicotine replacement therapy (bupropion and varenicline) is used for nicotine-related disorders, methadone and buprenorphine are used to treat opiate use disorders, and naltrexone, acamprosate, and disulfiram are used in alcohol-related disorders. There are no specific drugs to treat cannabis- and cocaine-related disorders. Although these treatments show some efficacy, the pharmacological interventions available are not effective for many patients and have key limitations such as daily dosing, treatment adherence, and adverse effects [1, 4, 5]. Therefore, it is necessary to explore new interventions that may be more effective.

Classic hallucinogens or serotonergic psychedelics such as ayahuasca, *N,N*-dimethyltryptamine (DMT), psilocybin, and lysergic acid diethylamide (LSD) are drugs that produce alterations in perceptions, mood, and cognition, and share the same mechanism of action through agonism at cortical serotonin 5-HT_{2A} receptors [6, 7]. Agonism at cortical 5-HT_{2A} receptors is associated with increased neuroplasticity in brain areas related to emotional processing, introspection, and social cognition, and this seems to be especially relevant for the therapeutic effects of these drugs [7–9].

Previous studies with these drugs for the treatment of SUDs performed in the early 1960s and 1970s showed promising results [10]. A meta-analysis of six randomized controlled trials (involving a total of 536 subjects) conducted between 1966 and 1970 evaluating the efficacy of LSD in the treatment of alcoholism showed the treatment with a single dose of LSD significantly reduced alcohol misuse for up to three months [11]. More recently, the effects of psilocybin were evaluated in two open-label studies: one for tobacco [12, 13] and the other for alcohol [10, 14, 15] use disorders. In the first study, smoking abstinence was observed up to six months after psilocybin treatment, and abstinence after treatment was associated with mystical-type experiences [16]. Moreover, persisting positive changes such as increased esthetic appreciation, altruism, and pro-social behavior were also reported 30 months after psilocybin sessions. Perceived preparatory counseling, rapport with the therapeutic team, and engagement in the treatment were considered key factors in achieving abstinence [13]. In the second study, psilocybin treatment was associated with a significant reduction in drinking days up to eight months after the treatment [14]. Mystical-type experiences and intensity of subjective effects were associated with improvements [10, 14, 15]. Corroborating these clinical results, reductions in drug use in general [17, 18] and in tobacco use specifically [19] have also been

reported in online surveys among hallucinogens/psychedelics users of the general population (using both “full” [17, 19], and “micro” [18] doses). In the tobacco study, abstinence was mediated by spiritual experience, changed priorities/values, and improved emotional regulation [19].

Besides the promising results regarding drug use, these substances have also shown the potential to treat depression and anxiety symptoms, which often co-occur with SUDs [20]. Two randomized trials assessed the effects of psilocybin for anxiety and depression in patients with cancer [21, 22] and two open-label trials assessed the effects of psilocybin in patients with depression [23, 24]. In all studies, psilocybin treatment was associated with significant reductions in symptoms of depression and anxiety from three to six months after drug intake. Thus, these drugs could also target these symptoms in SUDs.

Another classic hallucinogen/psychedelic with possible therapeutic use in the treatment of SUDs is ayahuasca. Ayahuasca is a hallucinogenic/psychedelic botanical drink used for ritual and therapeutic purposes by indigenous tribes in countries such as Brazil, Peru, Colombia, and Ecuador and by syncretic religions such as *Santo Daime*, *União do Vegetal* (UDV), and *Barquinha* present in South America and worldwide [7]. The beverage is prepared by the decoction of *Banisteriopsis caapi* and *Psychotria viridis*. *B. caapi* stalks are rich in β -carbolines such as harmine, harmaline, and tetrahydroharmine (THH), and *P. viridis* leaves contains the psychedelic tryptamine DMT [25, 26]. When ingested orally, DMT does not promote psychedelic effects due to degradation by monoamine oxidase type A (MAO-A). However, the β -carbolines extracted from *B. caapi* act as selective and reversible MAO-A inhibitors, making it possible for DMT to reach the Central Nervous System [25].

Evidence from anthropological and observational studies suggests that ritual and religious ayahuasca use is associated with reductions in drug use and other drug-related problems [27–31]. In these studies, increases in social bonds by feeling part of a community and enduring participation in communitarian activities (which are often intertwined with enhanced spirituality) seem to be key factors involved in the therapeutic potential of ayahuasca.

A previous systematic review [32] of quantitative preclinical and human studies assessing the effects of ayahuasca and its alkaloids regarding drug use and other measures related to SUDs analyzed 10 studies: five preclinical [33–37] and five observational studies [38–42]. Four [33–36] of the preclinical studies used β -carbolines and only one [37] used ayahuasca.

In the preclinical studies, harmaline (10–40 mg/kg) reduced cocaine and morphine self-administration in rats [33], while harmine (5–10 mg/kg) reduced withdrawal symptoms caused by naloxone in rats [34] and psychostimulant-related behavior in planarians (0.01–10 μ M) [36].

These effects could be related to the increased dopamine efflux produced by harmine in the rat nucleus accumbens shell, which was enhanced by cocaine co-administration ($260 \pm 35\%$) [35]. Another preclinical study showed that ayahuasca (30–500 mg/kg) reduced alcohol-induced hyperlocomotion and sensitization in mice [37].

Regarding the observational studies previously described [32], a retrospective assessment of the mental health of 37 UDV adolescent members and 43 controls did not find differences in lifetime drug use, but UDV adolescents had less last-year and last-month alcohol use [39]. In a study assessing the psychosocial consequences related to continuous ritual ayahuasca among 107 ritual ayahuasca users compared with 115 non-ayahuasca users, significant reductions in alcohol consumption, drug use, and psychiatric symptoms were observed, which were sustained at a 12-month follow-up [40]. Furthermore, two studies among patients with SUDs that were treated in ritual/clinical settings using ayahuasca showed sustained reductions in drug use, impulsivity, and anxiety, with increased quality of life, spirituality, well-being, and neuropsychological improvements [41, 42].

Taken together, the studies previously reviewed [32] suggest that ayahuasca and its alkaloids were associated with reduced drug use in animal models and observational studies. Moreover, improvements in psychological (anxiety, impulsivity) and cognitive (quality of life, well-being) measures related to SUDs were also observed. These results are in line with a recent observational study among healthy ritual ayahuasca users that reported significant reductions in depression, improvements in quality of life, and less use of prescription medications when compared with ayahuasca-naïve controls [43]. Furthermore, these results are corroborated by open-label [44] and controlled [45] trials showing that a single dose of ayahuasca significantly reduced depressive symptoms in patients with treatment-resistant depression. Therefore, ayahuasca could also be beneficial in treating co-occurring psychological and cognitive problems in people with SUDs.

Considering the above-mentioned studies, we updated the previous systematic review [32] evaluating quantitative preclinical and human studies assessing the effects of ayahuasca and its alkaloids on drug use (primary outcome) and other psychological and cognitive measures (secondary outcomes) related to SUDs.

Methods

The data for the present systematic review were collected according to the guidelines for systematic reviews and meta-analysis (Preferred Reporting Items for Systematic Reviews and Meta-Analyses [PRISMA]) [46].

Search strategy

Electronic searches were performed for studies published in the period from 01/01/2016 to 11/08/2020 on the Pubmed, Psycinfo, Scielo, and LILACS databases. The following keywords were used: (ayahuasca OR dimethyltryptamine OR harmine OR tetrahydroharmine OR harmaline) AND (dependence OR addiction OR substance use disorder OR animals).

Selection criteria and study selection

The inclusion criteria were: (a) randomized and open-label clinical studies using ayahuasca or any of its alkaloids in the treatment of SUDs; (b) observational studies describing the use of ayahuasca or any of its alkaloids in the treatment of SUDs; (c) quantitative studies with a statistical analysis comparing two or more groups at the same time or at least two different time points in the same group; (d) studies that used standardized instruments to assess the pattern of use of associated substances or symptoms; (e) preclinical studies using ayahuasca or any of its alkaloids in experimental models of SUDs; (f) studies published between 2016 and 2020; and (g) studies published in English, Spanish or Portuguese. The exclusion criteria were studies used as results from the previous review [32].

Recorded variables, data extraction, and analysis

For preclinical studies, the following data were extracted: year of publication, study design, sample characteristics (types of animals used and sample size, weight, and average age of the animals), substances used (ayahuasca, β -carbolines, and DMT), and administration routes (intra-peritoneal or gavage). For human studies, the following data were extracted: year and country of publication, study design, sample characteristics (sample size, the gender of volunteers, average age), and ayahuasca dose/treatment.

Evaluation of selected studies

To assess the quality of preclinical studies, the guidelines and checklist of the Systematic Review Center for Laboratory animal Experimentation (SYRCLES) were used. This instrument was based on and adapted from the Cochrane Collaboration RoB Tool and aims to evaluate the methodological quality of studies of their risk of bias [47]. To assess the quality of human studies, the checklist and guidelines for studies from the National Institutes of Health (NIH) were used (<https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools>). Grading was done by verifying the presence or absence of the information required on the checklist for each article. A “0”

indicates the information was absent and “1” indicates the information was present. The overall score of each article was calculated by dividing the sum of points by the difference of the total number of points in each scale less the not applicable (NA) points.

Results

Selected studies

A flow diagram illustrating the different phases of the articles selected for the systematic review is presented in Fig. 1.

Identification of studies

The bibliographic search was performed in the Pubmed, Scielo, Psycinfo and LILACS databases from January 1, 2016 to November 8, 2020, with 259 papers found in the

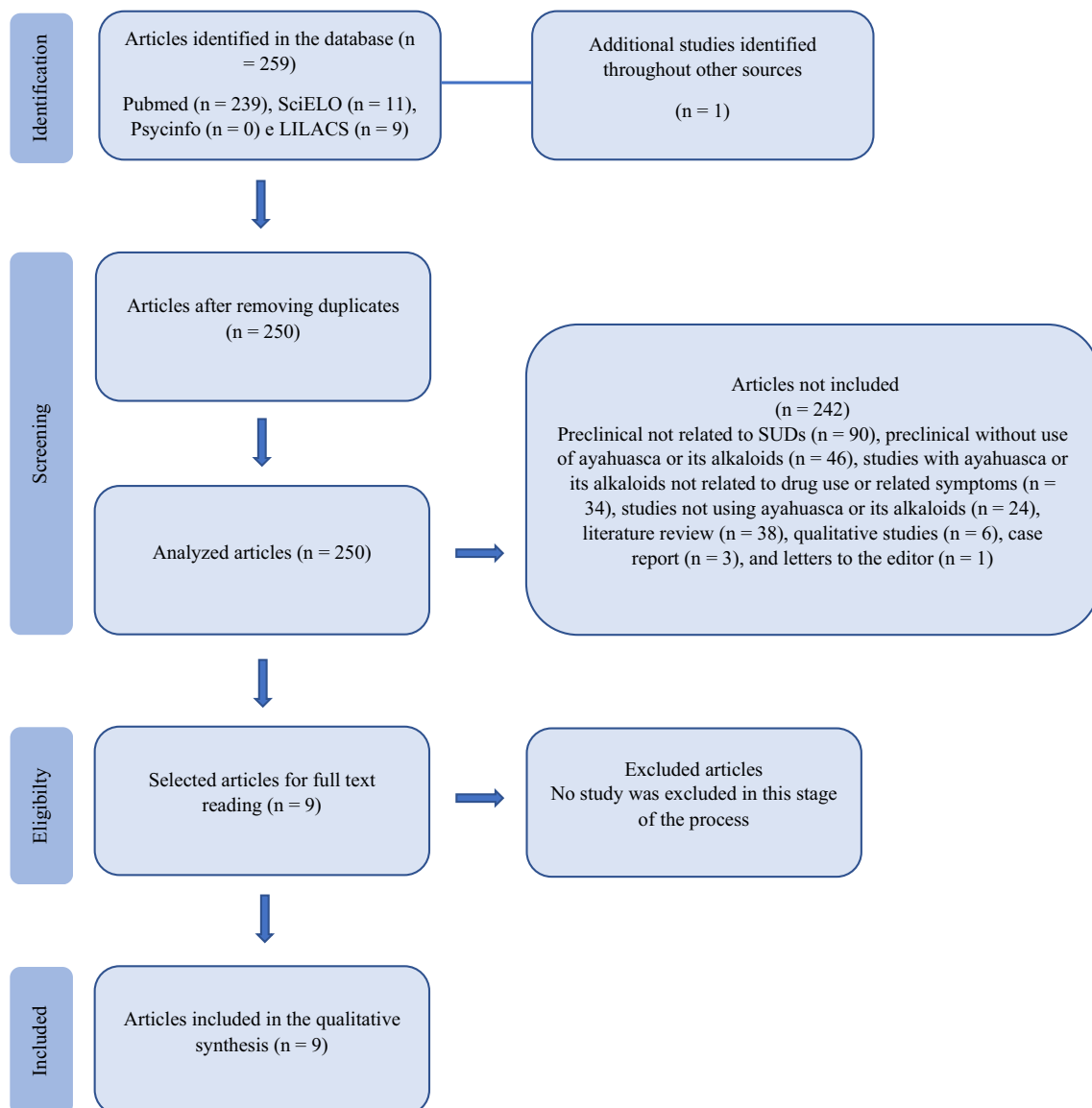


Fig. 1 Flowchart illustrating the stages of searching and selecting articles in the Pubmed, SciELO, Psycinfo and LILACS databases. SUDs: substance use disorders

first stage. After removing duplicates and reading the titles and abstracts of the articles found, eight references were selected for full reading considering the inclusion and exclusion criteria [48–55]. Another article was found while hand searching the bibliography of selected papers [56]. After full reading, no study was excluded, providing a total of 9 studies included in the systematic review. Of these, five were human observational studies [48–50, 52, 53] and four were preclinical studies [51, 54–56].

Human studies

The summary of the main information for all included human studies is described in Table 1. The first study found in the database search was conducted by Barbosa et al. [48] designed as an observational cross-sectional case–control study. This study investigated the psychological, neuropsychological functions, and substance use of 30 adult UDV members in the United States, compared to 27 non-ayahuasca adults from the same region. The American version of Nelson Adult Reading Test (AMNART), Stroop Color and Word Test, Conner’s Continuous Performance Test (CPT II), California Verbal Learning Test (CVLT), and the Trail Making Test (TMT) were used to evaluate the neuropsychological functions of the volunteers. To analyze the consumption of substances and psychological functions, researchers used the Medical Outcome Study Short Form (SF-36), the Big Five Inventory (BFI), Barratt Impulsiveness Scale (BIS-11), Profile of Mood States (POMS), and the Addiction Severity Index (ASI). Volunteers in the UDV group scored significantly lower in depression ($p=0.043$) and confusion ($p=0.032$) as indexed by the POMS scale, while also showing higher scores on the personality traits agreeableness ($p=0.028$) and openness ($p=0.037$) according to the BFI. The same group showed higher past use of alcohol ($p=0.007$) and cannabis ($p=0.001$) with lower recent alcohol consumption ($p<0.001$) as assessed by the ASI. The UDV group also scored better on memory vulnerability to proactive interference (disruption of new memory acquisition by interference from similar information that has been learned previously) as assessed by CVLT ($p=0.040$). Lastly, the UDV group showed higher scores on the quality life domain “role limitations due to physical health” as assessed by the SF-36 ($p=0.035$). It was also observed a correlation between prolonged consumption of ayahuasca and less alcohol consumption throughout life ($p=0.034$). AMNART, BIS-11, DUREL, DSES, CPT II, RCF, and Stroop Color and Word Test showed no significant difference. No adverse effects were reported [48].

The second study found was conducted by Lawn et al. [53] designed as an international cross-sectional online survey between November 2015 and February 2016. This study aimed to assess the well-being and problematic

alcohol consumption in a sample of 96,901 volunteers over 16 years old from different countries. The sample was comprised of three different groups: ayahuasca users ($n=527$), classic psychedelics (LSD and psilocybin mushrooms) users ($n=18,137$) and non-users ($n=78,236$). The Alcohol Urge Identification Test (AUDIT) and Personal Wellbeing Index (PWI) were used to assess the well-being and alcohol consumption of the volunteers. The ayahuasca group presented higher well-being compared to the group of classic psychedelics users and non-users as assessed by the PWI ($p<0.001$). The ayahuasca group also scored lower on the AUDIT for problematic alcohol consumption when compared to classic psychedelics users ($p=0.002$), while scoring higher on this scale when compared to non-users ($p=0.001$). There was a similar proportion of mental illness prevalence (people who reported having ever been diagnosed with a mental illness) between the ayahuasca (24%) and classic psychedelics users (22.4%), with ayahuasca users presenting a significantly higher proportion when compared with non-psychedelic users (18.8%, $p<0.001$). The “negative effects while high” were stronger for ayahuasca consumption when compared to LSD ($p<0.002$) and psychedelic mushrooms ($p<0.001$), but the authors did not distinguish in the survey which negative effects were felt by the volunteers [53].

Barbosa et al. [49] conducted a cross-sectional observational study to assess the impact of ayahuasca use on alcohol and tobacco consumption on UDV Brazilian members. The study consisted of a sample of adult members (18+ years old; $n=1947$ with an average age of 39.85 years) from the UDV from 10 different Brazilian states compared to a control group composed of adults (18–34 years old) from the general Brazilian population ($n=7939$). Sampling was acquired between March 2009 and August 2011. To analyze the consumption of alcohol and tobacco, the Brazilian version of the Substance Abuse and Mental Health Services questionnaire (SAMHSA) was used. The consumption of alcohol and tobacco was higher for the UDV group when compared to the general Brazilian population for the age groups of 25–34 and over 34 years old ($p<0.001$). However, this was not observed for the age group between 18 and 24 years ($p<0.001$). The UDV group had a lower prevalence of disorders related to alcohol and tobacco consumption in the 30 days prior to the data collection ($p<0.001$). UDV members with >3 years of association with the religion were less likely to have consumed alcohol and tobacco in the last 12 months, in the last 30 days or to meet the criteria for alcohol and tobacco consumption disorder when compared with members with <3 years ($p<0.005$) [49].

Berlowitz et al. [50] conducted a study with a prospective observational cohort design to assess the effects of ayahuasca on SUDs on males recently admitted to the Takiwasi therapeutic center in Peru for a 3–12-month treatment. The study focused on the severity of dependence, craving for

Table 1 Human studies of ayahuasca assessing drug use and related measures

References	Study design/location	Samples	Outcomes	Main results
Barbosa et al. [48]	Cross-sectional case-control study United States	30 adult UDV members compared to a matched control group of 27 people that had never consumed ayahuasca before	Primary: recent drug use (ASI) Secondary: psychiatric symptoms (POMS), health problems (SF-36), personality (BFI), impulsivity (BIS-11), religiosity/spirituality (DUREL, DSES), cognition (AMNART, CPT II, CVLT, ROCF, Stroop Color and Word Test)	The UDV group showed lower recent alcohol consumption (ASI), lower scores in depression and confusion (POMS), higher scores on personality (agreeableness and openness, BFI), and better memory (CVLT) and fewer health problems (SF-36). There was a correlation between prolonged consumption of ayahuasca and less alcohol consumption throughout life
Lawn et al. [53]	International online cross-sectional drugs survey Several countries	96,901 volunteers over 16 years old divided into three different groups: ayahuasca users, classic psychedelics users, and non-users	Primary: alcohol use (AUDIT) Secondary: well-being (PWI)	The ayahuasca group showed lower alcohol use (AUDIT) and higher well-being (PWI)
Barbosa et al. [49]	Cross-sectional study, Brazil	1947 UDV compared to normative data of the general Brazilian population as controls ($n = 7939$)	Primary: alcohol and tobacco use (SAMHSA)	The UDV group showed lower prevalence of disorders related to alcohol and tobacco in the last 30 days. Members with > 3 years of UDV association were less likely to consume alcohol and tobacco in the last 12 months, 30 days, or meet criteria for alcohol and tobacco disorders compared to members with < 3 years or less of association
Berlowitz et al. [50]	Prospective cohort study Peru	36 male volunteers from the Takiwasi Therapy Center diagnosed with drug abuse or dependence through DSM-IV criteria	Primary: drug use (ASI-5) Secondary: psychiatric symptoms (MINI), emotional distress (HADS), substance craving (CEQ), willingness to make life changes (RCQ), quality of life (WHOQOL-BREF)	There was a significant decrease for dependence severity, drug use, alcohol use, psychiatric status, and social/familial relationships (ASI), emotional distress (HADS) and substance craving (CEQ) comparing the participants before and after treatment. There was also a significant increase in the quality of life (WHOQOL-BREF). RCQ showed that most volunteers were willing to make life changes

Table 1 (continued)

References	Study design/location	Samples	Outcomes	Main results
Giovannetti et al. [56]	Prospective cohort study, Peru	31 male volunteers from the Takiwasi Therapy Center diagnosed with SUDs	Primary: anxiety (BAI) and depression (BDI). Secondary: satisfaction with treatment (CSQ), quality of life and spirituality (WHOQOL-SRPB; WHOQOL-BREF), treatment entry (TEQ)	Mean scores showed a significant reduction in depressive and anxiety symptoms (BDI and BAI) between before and after the treatment. There was a higher decrease in the scores of BAI and BDI in patients with opioid dependence. A correlation between the decrease in BAI and BDI scores with satisfaction of the treatment (CSQ) and increase in quality of life and spirituality was found (WHOQOL-SRPB and WHOQOL-BREF)
<i>AMNART</i> American Version of Adult Reading Test; <i>ASI</i> Addiction Severity Index; <i>ASI-5</i> Addiction Severity Index 5; <i>AUDIT</i> Alcohol use Disorder Identification Test; <i>BAI</i> Beck's Anxiety Inventory; <i>BDI</i> Beck's Depression Inventory; <i>BFI</i> Big Five Inventory; <i>BIS-11</i> Barratt Impulsiveness Scale; <i>CEQ</i> Craving Experience Questionnaire; <i>CPT II</i> Conner's Continuous Performance Test; <i>CSQ</i> Client Satisfaction Questionnaire; <i>CVLIT</i> California Verbal Learning Test; <i>DSES</i> Daily Spiritual Experience Scale; <i>DUREL</i> Duke University Religion Index; <i>HADS</i> Hospital Anxiety and Depression Scale; <i>MINI</i> Mini International Neuropsychiatric Interview; <i>PWI</i> Personal Wellbeing Index; <i>POMS</i> Profile of Mood States; <i>RCQ</i> Readiness to Change Questionnaire; <i>ROCF</i> Rey Osterrieth Complex Figure Test; <i>SAMHSA</i> Substance Abuse and Mental Health Service; <i>SF-36</i> Medical Outcome study Short Form; <i>SUDS</i> substance use disorders; <i>TCL-R</i> Temperament and Character Inventory-Revisited; <i>TEQ</i> Treatment Entry Questionnaire; <i>UDV</i> União do Vegetal; <i>WHOQOL</i> World Health Organization Quality of Life-BREF; <i>WHOQOL-SRPB</i> World Health Organization Quality of Life-SRPB				

substances, depressive and anxious symptoms, and quality of life of the volunteers ($n=36$ with an average age of 30.86 years old), one week after the final treatment day (average test–retest interval: 215.6 days ($SD=84.7$)). SUDs were diagnosed via DSM-IV criteria. Researchers used the Mini International Neuropsychiatric Interview (MINI) for psychiatric assessment, the Hospital Anxiety and Depression Scale (HADS) to assess emotional distress, the Addiction Severity Index (ASI-5) to analyze the severity of dependence symptoms, the Craving Experience Questionnaire (CEQ) to measure the frequency of cravings for substances, the World Health Organization Quality of Life-Bref (WHOQOL-BREF) to measure the quality of life, and the Readiness to Change Questionnaire (RCQ) to measure the motivation for changing lifestyle habits of the volunteers. There was a significant decrease for the dependence severity outcomes drug use ($p<0.001$), alcohol use ($p<0.001$), psychiatric status ($p<0.001$), and social/familial relationships ($p<0.001$) (as indexed by the ASI), emotional distress ($p<0.001$) (as indexed by the HADS), and substance craving ($p<0.001$) (as indexed by the CEQ) when comparing the participants from baseline to one week after the final treatment day. There was also a significant increase in the quality of life score as indexed by the WHOQOL-BREF ($p<0.001$). RCQ scale showed that most of the volunteers (61.1%) were willing to change life attitudes [50].

The last study found was conducted by Giovannetti et al. [52] with a prospective observational cohort design that assessed the effectiveness of ayahuasca on symptoms of depression and anxiety related to SUDs. The study was carried out with volunteers from the Takiwasi center in Peru admitted between January 2016 and 2018 (all males, $n=32$ with an average age of 32 years old). The Beck Depression Inventory (BDI) and Beck Anxiety Inventory (BAI) were used to assess depression and anxiety, the Treatment Entry Questionnaire (TEQ) was used to measure motivation to seek treatment, the Client Satisfaction Questionnaire (CSQ) was used to analyze satisfaction with care received, the WHOQOL-BREF were used to observe the subjects' quality of life, and the World Health Organization Quality of Life-Spirituality, Religiosity and Personal Beliefs subscale (WHOQOL-SRPB) was used to assess the volunteers' beliefs. The BDI and BAI scores showed a statistically significant reduction in depressive ($p<0.001$) and anxiety ($p<0.002$) symptoms when comparing before and after treatment (mean treatment duration 6.5 months). There was no statistical difference between the main drug abuse groups, but volunteers with opioid dependence showed a greater decrease in the scores of the BAI and BDI ($p<0.05$). A correlation was observed between the decrease in the scores of the BAI and BDI with the satisfaction of the treatment as assessed by the CSQ ($p<0.02$). An increase in the quality of life and spirituality scores was found through the

WHOQOL-SRBP and WHOQOL-BREF, correlating with the decrease in depression and anxiety scores ($p < 0.05$) [52].

The studies showed an average score of 60%, considered medium to high in terms of quality. In the studies by Barbosa et al. [48, 49] and Lawn et al. [53], the most common biases found were related to the size and power of the sample, blinding of the researchers during the analysis of results, and the data collection carried out in only one point of the experiment (questions 5, 11 and 12). The biases identified in the Berlowitz et al. [50] and Giovannetti et al. [52] studies were related to lack of blinding during data collection, absence of data collection at different time points, and relatively small sample sizes (questions 5, 8, and 11). The results for the quality assessment for all included human studies are presented in the Supplementary Material 1.

Preclinical studies

The summary of the main information for all included preclinical studies is described in Table 2. The first preclinical study used an animal model of amphetamine self-administration [56]. This study evaluated the effectiveness of ayahuasca (2 mL/kg) in modifying the behavioral effects (locomotion and anxiety) and self-consumption of amphetamine in rats. Animals were trained to ingest amphetamine for 13 days with or without ayahuasca. Twenty-four hours after the last day of treatment, locomotor activity and anxiety behaviors were evaluated using open-field arena (OP) and the elevated plus-maze (EPM). The rats were divided into four groups: control animals that received only water ($n = 15$), amphetamine group that received 0.6 mg/mL of amphetamine diluted in water ($n = 15$), ayahuasca group that received ayahuasca (AYA) through gavage ($n = 15$), and the amphetamine + ayahuasca group ($n = 15$). No statistical difference was observed in water consumption by animals in the control and ayahuasca groups. In the amphetamine group, water consumption was significantly higher ($p < 0.05$). Moreover, animals treated with ayahuasca consumed less water with amphetamine, suggesting that ayahuasca prevented the preference for amphetamine ($p < 0.05$). Animals that consumed amphetamine showed an increase in locomotor activity ($p < 0.05$) compared to the control group. The amphetamine + ayahuasca group did not show any significant difference in locomotor activity when compared to the control group, but presented less locomotor activity when compared with the amphetamine only group ($p < 0.05$). Animals that received amphetamine spent more time in the closed arms of the elevated plus-maze ($p < 0.05$), while animals that received ayahuasca + amphetamine spent less time in the closed arms ($p < 0.05$) [56].

Cata-Preta et al. [51] carried out a study in mice to observe the effects of ayahuasca (30, 100, and 300 mg/kg), *B. caapi* extract (EBc) (132, 440, and 1320 mg/kg)

and of *P. viridis* extract (EPv) (3.75, 12.5, and 37.5 mg/kg) in a Conditioned Place Preference (CPP) paradigm promoted by alcohol (1.8 g/kg). In this paradigm, a stimulus (in this case alcohol and ayahuasca) is associated with a particular area the animal has access to, and by measuring the amount of time the animal spends in that area, researchers can infer the animal's preference for the stimulus. Ayahuasca and EBc were administered via gavage, while EPv and alcohol were administered intraperitoneally. The results showed that ayahuasca at the dose of 100 mg/kg induced CPP ($p < 0.05$). Pretreatment with ayahuasca blocked the development of CPP induced by alcohol ($p < 0.05$). Ayahuasca in doses of 100 and 300 mg/kg decreased the alcohol-induced CPP effects when compared with the group treated with saline ($p < 0.01$). The 1320 mg/kg dose of EBc and 12.5 and 37.5 mg/kg doses of EPv blocked expression of alcohol-induced CPP in the alcohol-paired compartment ($p < 0.05$) [51].

Another preclinical study in rats was conducted by Nolli et al. [54] to assess voluntary alcohol consumption and the expression of cFos protein expression (a marker of strongly activated neurons) in brain areas related to substance consumption. The animals were divided in five groups: naltrexone ($n = 11$), ayahuasca 0.5 (0.26 mL/kg) (Aya0.5, $n = 10$), ayahuasca 1 (0.53 mL/kg) (Aya1, $n = 11$), ayahuasca 2 (1.06 mL/kg) (Aya2, $n = 12$) and controls ($n = 10$). During the experimental sessions, naltrexone and alcohol were administered intraperitoneally and ayahuasca was administered by gavage. To analyze alcohol consumption, the researchers placed the animals in separate cages with bottles containing water or alcohol and weighed the bottles before and after each experimental session. In addition, in each experimental session, the researchers exchanged the vials to avoid preference bias. The cFos expression analysis was made in brain areas associated with substance use and SUDs, including the medial orbital cortex (MO), lateral orbital cortex (LO), ventral orbital cortex (VO), nucleus accumbens (NAc), and the caudal putamen (CPu). The naltrexone group showed a significant reduction in alcohol intake when compared with the control group ($p < 0.05$). Alcohol significantly increased the expression of cFos in the MO in the control ($p < 0.0001$), naltrexone ($p < 0.05$), Aya1 ($p < 0.001$), and Aya2 ($p < 0.0001$) groups. This increase was also observed in the VO for Aya1 ($p < 0.05$) and in the LO for Aya2 ($p < 0.01$). An increase in cFos levels was also observed in the NAc of the Aya1, Aya2, and naltrexone groups ($p < 0.005$). cFos expression was significantly lower in the naltrexone ($p < 0.05$) and Aya0.5 ($p < 0.01$) groups in the MO compared with the control group, but only in the ayahuasca group levels achieved those of animals not exposed to ethanol. There was no reduction of alcohol consumption in the three ayahuasca treatment groups [54].

Table 2 Preclinical studies assessing the effects of ayahuasca on rodent models of drug use and related measures

Reference	Study design	Sample	Substances/doses	Routes of administration	Main results
Godinho et al. [56]	Self-ingestion of amphetamine, locomotor activity and anxiety behaviors	Animals: 60 Wistar rats with 22 days old. Divided in control ($n=15$), amphetamine ($n=15$), ayahuasca ($n=15$), and ayahuasca + amphetamine ($n=15$)	Amphetamine (0.6 mg/mL) and ayahuasca (2 mL/Kg)	Amphetamine diluted in water bottle and ayahuasca administered through gavage	Ayahuasca prevented preference for amphetamine reduced the increased locomotor activity produced by amphetamine. Amphetamine caused the animals to spend more time in the closed arms of the elevated plus-maze, with ayahuasca administration reversing this effect
Cata-Preta et al. [51]	CPP	Animals: Male swiss mice. Age: 3 months 5–7 animals allocated per cage	Ayahuasca in dose of 30, 100 and 300 mg/kg. <i>Banisteriopsis caapi</i> extract (EBc) in dose of 132, 440, 1320 mg/kg. <i>Psichotrya viridis</i> extract (EPv) in dose of 3.75, 12.5 and 37.5 mg/kg and alcohol at the dose of 1.8 g/kg	Ayahuasca, EBc and saline (as control for treatments) were administered through gavage. EPv, alcohol and saline (as control for alcohol CPP) were administered intraperitoneally	Ayahuasca (100 mg/kg) induced CPP, and pretreatment with ayahuasca blocked the development of CPP induced by alcohol. Ayahuasca (100 and 300 mg/kg) decreased the alcohol-induced CPP effects when compared with the group treated with saline. The EBc (1320 mg/kg) and EPv (12.5 and 37.5 mg/kg) blocked expression of alcohol-induced CPP in the alcohol-paired compartment
Nolli et al. [54]	Self-administration	Animals: 60 Male Wistar rats Weight: 250 g divided in control ($n=12$), naltrexone ($n=12$), ayahuasca0.5 ($n=12$), ayahuasca1x ($n=12$), and ayahuasca2x ($n=12$)	2 mg/kg naltrexone, 20% alcohol solution, and ayahuasca (0.26 mL/kg or Aya0.5, 0.53 mL/kg or Aya1, and 1.06 mL/kg or Aya2	Alcohol was self-administered orally, naltrexone was administered intraperitoneally, and ayahuasca was administered through gavage	Naltrexone significantly reduced alcohol intake. There was an increase in cFos expression in the MO in the control, naltrexone, Aya1, and Aya2 groups by alcohol; in the VO in the Aya1 group; and in the LO in the Aya2 group. There was also an increase in cFos expression in the NAc in the Aya1, Aya2 and naltrexone groups, with a remission of cFos expression in naltrexone and Aya0.5 groups

Table 2 (continued)

Reference	Study design	Sample	Substances/doses	Routes of administration	Main results
Reis et al. [55]	CPP	Animals: Male Swiss mice Age: 3 months 5–7 animals allocated per cage	10 mg/kg methylphenidate and 10 mL/kg ayahuasca. Both drugs were administered diluted in 0.9% saline as vehicle solution	Methylphenidate and saline were administered intraperitoneally and ayahuasca was administered through gavage	Both substances induced CPP behavior, which was greater for methylphenidate. Methylphenidate decreased levels of Fos in Cg1 PrL and OFC-LO in the post-conditioning test. The group exposed to methylphenidate and treated with ayahuasca showed reduction in CPP behavior. Ayahuasca prevented an increase of Fos levels in the NAc shell and NAc core after re-exposure to methylphenidate and the remission in Fos expression in Cg1, PrL and OFC-LO

Cg1 cingulate cortex area-1; *CPP* conditioned place preference; *LO* lateral orbital cortex; *MO* medial orbital cortex; *NAc* core nucleus accumbens core; *NAc shell* nucleus accumbens shell; *OFC-LO* orbitofrontal cortex-lateral orbital cortex; *PrL* prelimbic cortex; *VO* orbital lateral cortex

The study conducted by Reis et al. [55] investigated the effects of ayahuasca treatment (100 mg/kg administered via gavage) on the reward properties of methylphenidate (10 mg/kg administered intraperitoneally) in mice using the CPP paradigm. The study also assessed the expression of Fos protein in limbic regions associated with the behavioral effects of drugs of abuse (cingulate cortex, pre-limbic cortex, infralimbic cortex, frontal orbit cortex, shell and center of the NAc, ventral tegmental area, dorsal striatum, and basolateral amygdala). Both substances induced CPP behavior, but this effect was greater for methylphenidate than for ayahuasca ($p < 0.05$). When testing the effects of treatment with ayahuasca or methylphenidate on the CPP paradigm, methylphenidate significantly decreased the expression of Fos protein in the cingulate cortex area-1 (Cg1) ($p < 0.02$), prelimbic cortex (PrL) ($p < 0.0001$), and frontal orbital cortex-orbitolateral area (OFC-LO) ($p < 0.002$) in the post-conditioning test compared with the preconditioning test. Conditioning with methylphenidate, but not ayahuasca, also increased Fos expression in the NAc shell ($p = 0.0029$) and dorsal striatum ($p = 0.0017$). Conditioning with both ayahuasca and methylphenidate increased Fos expression in the infralimbic cortex ($p < 0.0001$) and the NAc core ($p = 0.0001$). Conditioning only with ayahuasca decreased Fos expression in the basolateral amygdala ($p < 0.0001$). Fos expression was not altered in the ventral tegmental area by ayahuasca or methylphenidate. When testing the effects of ayahuasca treatment on the reinstatement of methylphenidate-induced CPP, the group exposed to methylphenidate and treated with ayahuasca showed reduced CPP behavior when compared with the non-treated group ($p = 0.01$). Ayahuasca treatment restored Fos expression back to baseline (preconditioning test) levels in the Cg1 ($p < 0.0001$), PrL ($p = 0.0001$), and OFC-LO ($p = 0.0009$). After re-exposure to methylphenidate, this effect of was maintained in all three brain regions. Moreover, ayahuasca treatment prevented an increase of Fos levels in the NAc shell ($p < 0.0008$) and core ($p < 0.0001$) [55].

The studies showed an average score of 39%, considered low to moderate quality. The biases most prevalent according to the quality instrument were the absence of randomization in the group allocations of animals (question 1), randomization of the interventions administered to animals (questions 4 and 5), absence of animals' data collection randomization, and absence of researchers' blinding to treatments (questions 6 and 7). The quality assessment for all included preclinical studies is presented in the Supplementary Material 2.

Discussion

In the present systematic review, all preclinical studies reported mostly positive results, including reductions in amphetamine self-administration and anxiety [56] and in

alcohol- and methylphenidate-induced CPP [51, 55], and inhibition of some alcohol- and methylphenidate-induced modifications of cFos/Fos protein expression [54, 55]. The quality of the studies was low to moderate. On the other hand, ayahuasca also induced CPP in some of the dosages tested [51, 55], and did not reduce alcohol consumption [54]. Observational studies were conducted among ritual ayahuasca users [48, 49], patients with SUDs [50, 52], and classic hallucinogen/psychedelic users of the general population [53]. Overall, results were also mostly positive, including reductions in drug use [48–50, 53] and in anxiety and depression [48, 52], personality changes, and increases in quality of life and well-being [50, 52, 53]. Contrastingly, some studies reported greater lifetime substance use and mental health disorders in ayahuasca users compared to controls [49, 53]. The quality of the articles was medium to high.

Our results were similar to those described in our previous systematic review by Nunes et al. [32], but a significant difference between reviews was observed regarding preclinical studies: in the first review most studies used isolated β -carbolines [33–36] and only one study used ayahuasca [37]. As in the present review, results were mostly positive, including reductions in self-administration of cocaine and morphine [33], opioid withdrawal [34], alcohol-induced behavior [37] in rodents, and reduced C-shape responses to psychostimulants in planarians [36]. As in the present review, observational studies in the previous review were also conducted among ritual ayahuasca users [38–40] and patients with SUDs [41, 42] with positive results (i.e., reductions in drug use and psychiatric symptoms with increases in quality of life).

Regarding observational studies, in the present review two observational studies showed that alcohol and tobacco consumption by UDV members was higher throughout life when compared to controls, but recent consumption (last 30 days and 12 months) of these substances and related disorders was lower [48, 49]. This corroborates a previous study that showed higher consumption of cocaine and tobacco throughout life by UDV teenagers compared to a control group, with decreased consumption of substances such as amphetamines, alcohol, and tobacco in the period of the last 30 days–12 months at the time they participated in the research [39]. In comparison with these results, an open-label study with 10 volunteers carried out with psilocybin showed remission in heavy and daily alcohol consumption after the administration of the drug during an eight-month follow-up [14]. The researchers correlated the effects of remission on alcohol consumption with the quality of the experience promoted by psilocybin. However, despite the promising results from this study, it is important to note its limitations, such as the lack of a control group, blinding of the researchers and sample size [14]. Moreover, as with

the study with UDV teenagers, despite the promising data collected by Barbosa et al. [48, 49] this evidence needs to be observed with caution considering that the studies are observational and that the data collected were from members of a religion which uses ayahuasca as a sacrament (possible positive bias and difficulty in elucidating whether the beneficial effects are caused by ayahuasca or the environment in which the volunteers were inserted).

Lawn et al. [53] reported a decrease in problematic alcohol consumption in ayahuasca users in the previous year when compared with users of other serotonergic psychedelics (psilocybin and LSD), and greater alcohol intake when compared to the group of volunteers who did not consume any of the psychedelics. Regarding the well-being of the volunteers, ayahuasca users showed greater well-being when compared to both groups. Similar data were found in a survey among recreational users of hallucinogens/psychedelics carried out by Garcia-Romeu et al. [17], who reported a decrease in the consumption of opioids, stimulants, and cannabis after using LSD and psilocybin. In addition, both studies corroborate data found in the online cross-sectional study by Johnson et al. [19], which showed abstinence or remission in the use of tobacco by study volunteers after consuming serotonergic psychedelics.

Despite the promising data observed by these researchers, it is important to note some limitations present in these studies, such as their observational design, the retrospective nature of the data, and possible omissions in the amount of alcohol or tobacco reported to have been consumed by the volunteers. Also, the study samples were self-selected, so it is not a representative sample of drug users of the global population. Moreover, it is important to notice that the lifetime substance use of the volunteers from some studies was greater than that of controls [49, 53]. Nevertheless, a possible explanation for this fact is that ayahuasca is traditionally used with healing and medicinal purposes (different from usual LSD and psilocybin use), so people who have substance use problems may search specifically for ayahuasca in search of treatment. In fact, the results from the studies reviewed point in this direction, where recent (months to a year prior to data collection) drug use is reduced when comparing frequent ayahuasca users with other psychedelic users and the general population, even in cases where ayahuasca users showed greater lifetime substance use [49, 53]. The same reasoning may apply to the greater prevalence of mental health disorders in the ayahuasca group when compared with controls [53].

Two studies in the present review described patients diagnosed with SUDs that were treated at the Takiwasi center in Peru, a healing center that uses ayahuasca and other Amazonian plants and healing methods in a prolonged treatment program [50, 52]. Results of both studies showed a significant reduction of drug use by these patients, especially

for alcohol and opiates. Moreover, there was a significant improvement in the quality of life, craving for substances, emotional stress, and a reduction in the mean scores of depression and anxiety symptoms [50, 52]. These results corroborate data from previous studies performed in healing contexts [41, 42], where a significant reduction in the consumption of tobacco, alcohol and cocaine was observed. However, besides being observational, all these studies were performed in the context of multiple healing practices that used other therapeutic techniques in the mid to long term, making it difficult to separate the effects caused by therapy from those of ayahuasca.

Regarding personality measures assessed in the reviewed studies, UDV members scored higher on the agreeableness and openness traits of the BFI [48]. It is difficult to establish the causative role that ayahuasca could have on the personality, since these traits could already be present. For instance, agreeableness, openness, self-transcendence, compassion, and spiritual acceptance could render those participants more prone to use hallucinogens/psychedelics. In any case, the role of those substances in the modulation of personality traits is still poorly investigated [59].

Regarding preclinical studies, in the present review, two rodent models showed that ayahuasca reduced CPP for alcohol [51] and methylphenidate [55]. In the alcohol study, pretreatment with ayahuasca (100 and 300 mg/kg), the *B. caapi* (1320 mg/kg), and *P. viridis* (12.5 and 37.5 mg/kg) extracts decreased alcohol-induced CPP, while the 30 mg/kg ayahuasca dose did not produce significant effects and the 100 mg/kg dose also induced CPP [51]. Interestingly, in the methylphenidate study, the 10 mg/kg ayahuasca dose induced CPP (which was greater for methylphenidate) and reduced the methylphenidate-induced CPP [55]. These results corroborate findings from a previous study in which a reduction in alcohol-induced behavior was observed when animals were treated with ayahuasca doses of 300–500 mg/kg [37]. These results indicate the possible antiaddictive properties of ayahuasca, since it diminished the preference of the animals to areas associated with alcohol and methylphenidate consumption. The differences in significant dosages observed between studies could be related to study design and drugs evaluated (alcohol vs. methylphenidate). Ayahuasca-induced CPP could be due to increased dopamine flow in the Nac produced by harmine [35]. Together, these results suggest that ayahuasca and its alkaloids could have beneficial effects in alcohol- and psychostimulant-related disorders.

Similar effects were seen in other preclinical trials using rats to assess the effectiveness of harmine in self-consumption of morphine and cocaine [33], withdrawal symptoms from morphine [34], and efficacy of harmine in inhibiting psychostimulant responses (cocaine, methamphetamine, mephedrone, and nicotine) in planarians [36]. The reduced

consumption of these substances suggest that the administration of ayahuasca or harmine seems to decrease the reinforcement effects caused by them [33, 34, 36, 56].

Moreover, ayahuasca generally blocked the modifications of cFos/Fos protein expression (a marker of strongly activated neurons) produced by alcohol and psychostimulants [54, 55], and reduced amphetamine self-administration and anxiety and locomotor activity [56]. Nolli et al. [54] showed that alcohol increased the expression of the cFos protein in brain regions related to SUDs in animals, such as the orbital medial cortex, orbital ventral cortex, lateral orbital cortex, and the nucleus accumbens of animals. Ayahuasca also increased cFos expression in those areas, but animals exposed to alcohol and treated with ayahuasca showed decreased cFos expression. These results suggest a protective effect of ayahuasca in this region, at least at low levels, since higher ayahuasca doses did not show any impact on cFos levels caused by alcohol exposure. In the study by Reis et al. [55], ayahuasca blocked most of the changes in Fos expression induced by conditioning with methylphenidate and/or re-exposure to this drug. Moreover, while ayahuasca had limited effects on Fos expression, methylphenidate modified it in several brain regions associated with the behavioral effects of drugs of abuse. This result indicates that the rewarding properties of methylphenidate are more prominent. The above-mentioned effects could be due to the effects of ayahuasca on dopaminergic, glutamatergic, and serotonergic neurotransmission in those areas [57, 58]. Future studies should explore other dosages and drugs to better understand the effects of ayahuasca on cFos/Fos expression.

SUDs are complex and multifactorial disorders that need a treatment model which targets not only the individual at a bio-psychological level, but also his/her social network, attitudes/behaviors, and family. The possible mechanisms of the therapeutic effects of ayahuasca in SUDs are not fully understood. From a neurochemical perspective, agonism at cortical 5-HT_{2A} expressed in brain areas related to emotional processing, introspection, and social cognition, together with increased neuroplasticity, are possible mechanisms [7–9]. The effects of these drugs in limbic and frontal structures could be associated with the increases in cFos/Fos expression observed in rodent models of SUDs [54, 55]. These biochemical alterations could be responsible for improvements in emotional regulation, cognitive flexibility, and disruption of repetitive and negative thoughts.

Moreover, the evidence on the positive effects of harmine, harmaline, and ayahuasca in preclinical models from planarians to rodents using drugs as diverse as alcohol and amphetamine (reviewed by Nunes et al. [32] and in the present review) suggest a strong biological basis for the therapeutic effects of these substances in SUDs. This perspective is further reinforced by a recent study in mice showing that

LSD reduced alcohol consumption and preference, with reductions sustained for 46 days [60]. However, a recent study in rats failed to find long-lasting effects of high (two doses one week apart) and micro (two doses per week for four-weeks) doses of LSD and psilocybin in alcohol relapse, although they observed a short-lasting anti-relapse effect with sub-chronic treatment (five doses across three days) with psilocybin [61]. To the best of our knowledge, these are apparently the only published work on the effects of LSD/psilocybin on the self-administration of abused substances in animals. However, rodent studies with administration of other 5-HT_{2A} agonists also reported reductions of drug (e.g., alcohol, opiates, nicotine) preference and self-administration [62–65]. Nevertheless, although animal models related to SUDs are promising, it is not clear to which extent they are able to model the therapeutic potential of hallucinogens/psychedelic in humans.

At the psychological level, mystical-type experiences and the intensity of the subjective effects were associated with improvements in open-label clinical trials [12, 14–16], while spiritual experience, changed priorities/values, and improved emotional regulation during tobacco abstinence among hallucinogens/psychedelics users of the general population in an online survey have also been reported [19]. Perceived preparatory counseling, rapport with the therapeutic team, and engagement in the treatment were key factors in achieving drug abstinence after psilocybin sessions, which were accompanied by persisting (30 months after psilocybin sessions) positive changes including increased esthetic appreciation, altruism, and pro-social behavior [13]. Furthermore, positive changes in personality were also verified in observational studies [48]. Importantly, although these are complex psychological processes, human studies are showing increasing evidence that they are 5-HT_{2A}-mediated [57, 59].

In the culture/social level, observational studies conducted since the early 1990s suggest participation in syncretic ayahuasca religions and treatment of SUDs with ayahuasca in specialized clinics are associated with a reduction in drug misuse and related problems [27–29, 31]. These studies suggest that increases in social bonds and spirituality are important mediators of the therapeutic effects of ayahuasca. More recently, we reported that Spanish ritual ayahuasca users had higher scores on psychosocial adjustment, stress coping strategies, and indicators of social support compared with the general population [66]. These factors could be important mediators of improvement in SUDs. More studies are needed to clarify the mechanisms involved in the therapeutic effects of classic hallucinogens/serotonergic psychedelics, at the individual and social level.

Regarding the limitations of this review, there are important points that need to be mentioned. A total of nine studies were found, which is a small sample. Four studies were pre-clinical trials with low to moderate quality. Most limitations

were related to bias during the allocation of animals to the designated groups (selection bias), absence of randomization of substance administration to animals, blinding of the researchers themselves (performance bias), and the absence of randomization of the animals chosen for data analysis (detection bias). Moreover, it is difficult to translate/extrapolate the data found in these studies to humans. The other five studies were observational studies with medium to high quality. However, results need to be carefully evaluated since causality cannot be established. Furthermore, two studies [50, 52] were carried without a control group and in a therapeutic center with multiple therapeutic techniques applied together with ayahuasca. Thus, it is difficult to attribute any positive results to a single event. Furthermore, other studies were conducted among members of religions who use ayahuasca as a sacrament. Therefore, it is difficult to infer if the therapeutic effects are produced by ayahuasca or the context. Further limitations include small samples, lack of prolonged follow-ups, and wide variation in ayahuasca doses and frequency of use.

Conclusions

Preclinical and observational studies suggest that the ayahuasca may have therapeutic effects in the treatment of SUDs. Although the majority of the volunteers who composed the samples of the studies had a higher consumption of substances (mainly alcohol and tobacco) relative to the control groups, this consumption was lower in the last year prior to their participation in the studies. In addition, ayahuasca also increased quality of life and reduced depressive and anxiety symptoms. In studies with animal models, the data suggest that ayahuasca can promote CPP blocking induced by alcohol and promote remission in the self-consumption of amphetamine diluted in water. Preclinical data also suggest that ayahuasca promotes remission of cFos protein expression in brain regions related to drug dependence.

Despite the promising effects reported in this systematic review, it is important to mention that the studies have limitations, such as preclinical studies, observational designs, and small sample sizes. Moreover, some of the data were collected from volunteers who used ayahuasca as a sacrament, which could be biased. Because of this, greater methodological rigor is required to avoid the risk of bias in future research with ayahuasca or other serotonergic hallucinogens. Finally, no clinical trials were found in the present review. Future randomized, placebo-controlled clinical trials are needed to replicate the data found in this systematic review.

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Declarations

Conflicts of interest The authors declare no conflict of interest.

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